

NET731 – Cloud Networking

Week 9 Practical: Final Documentation

1 – 7 May 2026

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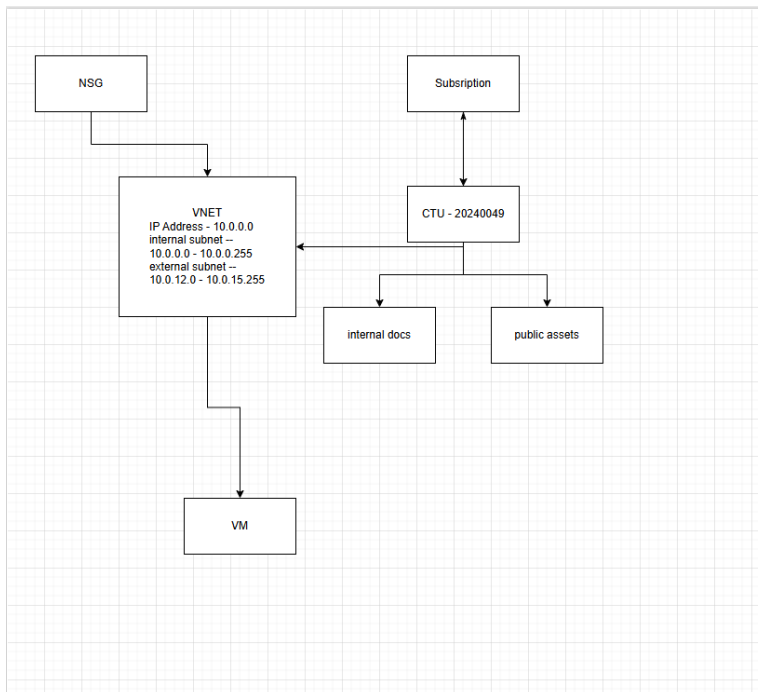
Student Number: 20240049

Overview

This document provides a full overview of the Azure cloud environment built during the NET731 practical project. The report includes the architecture design, resource configurations, security setup, monitoring configuration, and cost management summary.

Task 1: Architecture Diagram

The architecture diagram below represents the deployed Azure environment, including networking, security, monitoring, and compute resources.



Architecture Components Included

- Resource Groups
- Virtual Networks and Subnets
- Network Security Groups (NSGs)
- Virtual Machines
- Storage Accounts
- Azure Monitor
- Log Analytics Workspace
- Traffic Flow Between Resources

Task 2: Azure Environment Documentation

Section 1: Resources and Configurations

Resource Name	Type	Resource Group	Region	SKU/Tier	Purpose
Week8-VM	Virtual Machine	Week8-RG	South Africa North	B1s	Testing and networking practicals
Week8-VNet	Virtual Network	Week8-RG	South Africa North	Standard	Network communication
Week8-Storage	Storage Account	Week8-RG	South Africa North	Standard LRS	Storage and diagnostics
Week8-NSG	Network Security Group	Week8-RG	South Africa North	Standard	Traffic filtering
Week8-LAW	Log Analytics Workspace	Week8-RG	South Africa North	PerGB2018	Centralised logging

Each resource was configured to support networking, monitoring, security, and cloud management requirements for the project environment.

Resource Screenshots

Home

CTU-20240049-RG

Generate Bicep code to duplicate this resource group

Export resource groups using Bicep or Terraform

How to manage changes with deployment tools?

Search

Create

Manage view

Delete resource group

Refresh

Export to CSV

Open query

Assign tags

Move

Delete

Export template

Open in mobile

Add to service group

Group by none

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Cost Management

Monitoring

Insights (preview)

Alerts

Metrics

Diagnostic settings

Logs

Advisor recommendations

Workbooks

Automation

Help

Essentials

Subscription (move)

Subscription ID

Tags (edit)

Resources

Recommendations

Deployments

Location

JSON View

Filter for any field...

Type equals all

Location equals all

Add filter

Name	Type	Location
20240409storage	Storage account	Italy North
LogforVM	Log Analytics workspace	Italy North
MSVM-uuenorth-vm-week6	Data collection rule	UAE North
MSVM-uuenorth-week6-vm	Data collection rule	UAE North
MSVMOTel-uuenorth-vm-week6	Data collection rule	UAE North
NSG-20240049	Network security group	Italy North
vm-week6	Virtual machine	UAE North
vm-week6-ip	Public IP address	UAE North
vm-week6-nsg	Network security group	UAE North
vm-week6408_z3	Network interface	UAE North
vm-week6_key	SSH key	UAE North
vm-week6_OsDisk_1_e2f689c2139e4a67a23e3eb5e2fa1	Disk	UAE North
week6-vm	Virtual machine	UAE North

Showing 1 - 15 of 15. Display count: auto

Give feedback

vm-week6

Help me copy this VM in any region

Manage this VM with Azure CLI

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Monitor

Resource visualizer

Connect

Networking

Network settings

Load balancing

Application security groups

Network manager

Settings

Availability + scale

Security

Backup + disaster recovery

Operations

Auto-shutdown

Run command

Updates

Health monitoring

Configuration

Advisor (1 of 9): Enable Trusted Launch foundational excellence, and modern security for Existing Generation 2 VM(s) →

Help me copy this VM in any region

Connect

Start

Restart

Stop

Hibernate

Capture

Delete

Refresh

Scale

Open in mobile

Feedback

CLI / PS

JSON View

Essentials

Resource group (move)

Status

Location

Subscription (move)

Subscription ID

Availability zone

Operating system

Size

Primary NIC public IP

Virtual network/subnet

DNS name

Health state

Time created

Tags (edit)

Add tags

Properties

Monitoring

Capabilities (7)

Recommendations (9)

Tutorials

Virtual machine

Computer name

Operating system

VM generation

VM architecture

Agent status

Agent version

Hibernation

Host group

Host

Proximity placement group

Colocation status

Capacity reservation group

Disk controller type

Networking

Public IP address

Public IP address (IPv6)

Private IP address

Private IP address (IPv6)

Virtual network/subnet

DNS name

Size

Size

Virtual machine

Computer name

Operating system

VM generation

VM architecture

Agent status

Agent version

Hibernation

Host group

Host

Proximity placement group

Colocation status

Capacity reservation group

Disk controller type

vm-week6

Linux (ubuntu 24.04)

V2

x64

Ready

2.15.1.3

Disabled

-

-

-

N/A

-

SCSI

Networking

Public IP address

Public IP address (IPv6)

Private IP address

Private IP address (IPv6)

Virtual network/subnet

DNS name

Size

Size

vCPUs

RAM

Threads per core

20.174.3.158 (Network interface vm-week6408_z3)

1 associated public IPs

-

10.0.0.5

week6-vm-vnet/default

Configure

Standard D2s v3

2

8 GiB

2

Home

20240049storage

Storage account

Upload

Open in Explorer

Delete

Move

Refresh

Open in mobile

CL / PS

Feedback

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

> Data storage

> Security + networking

> Data management

> Settings

> Monitoring

> Automation

> Help

How can I make my storage account more resilient?

Is this storage account vulnerable?

Why can't this storage account be accessed?

Essentials

Resource group (myrg)

Location

Subscription (mysub)

Subscription ID

Disk state

Tags (edit)

CU / PS

Feedback

Performance

Standard

Replication

Locally-redundant storage (LRS)

Account kind

StorageV2 (general purpose v2)

Provisioning state

Succeeded

Created

2026/05/03, 13:49:09

Properties

Monitoring

Capabilities (7)

Recommendations (3)

Tutorials

Tools + SDKs

Blob service

Hierarchical namespace

Disabled

Default access tier

Hot

Blob anonymous access

Disabled

Blob soft delete

Enabled (7 days)

Container soft delete

Enabled (7 days)

Versioning

Disabled

Change feed

Disabled

NFS v3

Disabled

Allow cross-tenant replication

Disabled

Storage tasks assignments

None

File service

Identity-based access

Not configured

Default share-level permissions

Disabled

Soft delete

Enabled (7 days)

Require Encryption in Transit for SMB (Security)

Enabled

Security

Require secure transfer for REST API operations

Enabled

Storage account key access

Enabled

Minimum TLS version

Version 1.2

Infrastructure encryption

Disabled

Networking

Public network access

Enabled

Public network access scope

Enable from all networks

Private endpoint connections

0

Network routing

Microsoft network routing

Endpoint type

Standard

Rule Name	Direction	Port	Protocol	Source	Purpose
Allow-RDP	Inbound	3389	TCP	My IP	Remote desktop access
Allow-HTTP	Inbound	80	TCP	Any	Allow web traffic

NSG rules were configured to control inbound and outbound traffic based on the principle of least privilege.

RBAC and Identity Management

Role-Based Access Control (RBAC) was used to manage access to Azure resources. Contributor and Reader roles were assigned where required to ensure proper access control.

Defender for Cloud Recommendations

Microsoft Defender for Cloud recommendations were reviewed to identify security risks. Recommendations related to monitoring, endpoint protection, and networking were addressed where possible.

Security Screenshots

The screenshot displays the 'Role assignments' page in the Azure portal for user 'Ian Strauss' at the scope 'NSG-20240049'. The page shows five current role assignments. The left sidebar contains navigation links for Home, Overview, Activity log, Access control, Tags, Diagnose, Resource group, Settings, Inbound, Outbound, Network, Subnet, Properties, and Lock.

Role	Description	Scope	Group assignment	Condition
Contributor	Grants full access to manage all r...	Resource group (Inher...	--	None
Owner	Grants full access to manage all r...	Subscription (Inherited)	--	None
Owner	Grants full access to manage all r...	Subscription (Inherited)	--	None
Reader	View all resources, but does not ...	Resource group (Inher...	--	None
Storage Blob Data Contr	Allows for read, write and delete ...	Resource group (Inher...	--	None

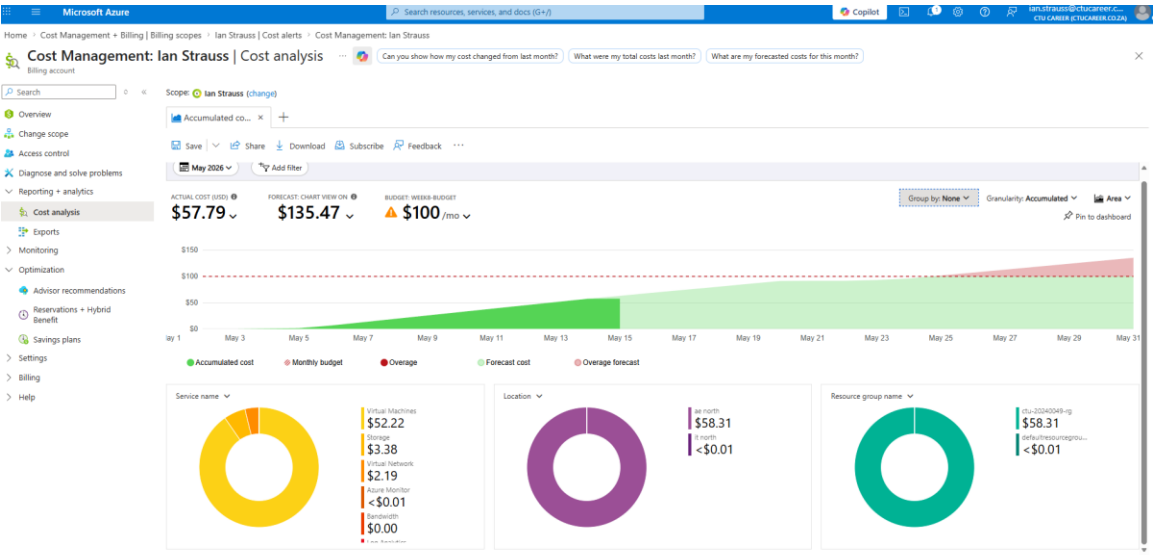
Deny assignments (0)

Section 3: Monitoring and Alerting

- CPU usage monitoring enabled on the virtual machine
- Disk and network metrics monitored through Azure Monitor
- Budget alerts configured in Azure Cost Management
- Log Analytics Workspace configured for log collection

Azure Monitor and Log Analytics Workspace were used to monitor performance, resource usage, and system logs throughout the project.

Monitoring Screenshots



Section 4: Cost Summary

Resource Type	Estimated Cost	Notes
Virtual Machines	\$25	Highest compute cost
Storage Accounts	\$10	Disk storage and diagnostics
Networking	\$5	Public IP and bandwidth

The virtual machine was identified as the highest-cost resource. The VM was stopped and deallocated when not in use to reduce compute costs.

Based on the current usage trend, the projected total spend remains within the allocated \$100 budget.

Task 3: GitHub Repository Structure

```
NET731-Project/
|
├── docs/
│   ├── diagrams/
│   │   └── architecture-diagram.png
│   ├── azure-environment.md
│   └── screenshots/
└── cost-management/
```

| — cost-report.md
| — README.md

Submission Checklist

- ☒ Architecture diagram exported and saved
- ☒ azure-environment.md completed
- ☒ Screenshots uploaded
- ☒ README.md updated
- ☒ Files committed and pushed to GitHub
- ☒ Repository shared or public